

A COUNTEREXAMPLE TO THE JAHANBEKAM–WEST MATCHING CONJECTURE

ABSTRACT. Let $m(n, t)$ be the maximum number of colors in an edge-coloring of K_n containing no t edge-disjoint rainbow perfect matchings. Jahanbekam and West conjectured that

$$m(n, t) = \binom{n-2}{2} + t$$

for even $n > 4$ and $2 \leq t < n$. For every even $n \geq 6$, we construct a coloring of K_n with $\binom{n-2}{2} + n$ colors and no $n-1$ edge-disjoint rainbow perfect matchings. Thus the conjecture fails for $t = n-1$.

A subgraph of an edge-colored graph is *rainbow* if its edges have pairwise distinct colors. For even n and $2 \leq t < n$, let $m(n, t)$ denote the maximum number of colors in an edge-coloring of K_n containing no t edge-disjoint rainbow perfect matchings. Jahanbekam and West [1, Theorem 1.1] proved that

$$m(n, t) \geq \binom{n-2}{2} + t,$$

with equality when $n > 4t + 10$. Immediately after that theorem they write “We believe that equality holds whenever $n > t \geq 2$ if $n > 4$; note that $m(n, t) = \binom{n-2}{2} + t + 1$ when $n = 4$ ”, and at the start of [1, Section 4], having described the construction that gives the lower bound, “We conjecture that the construction is optimal for $n > t$ ”. The conjectured range thus includes $t = n-1$, whereas the range $n > 4t + 10$ in which equality is proved never does, since $t = n-1$ there would require $n > 4n + 6$.

The exception at $n = 4$ recorded in the first of these sentences is the seed of the coloring below. For $n = 4$ the set B defined in the proof is itself a perfect matching of K_4 ; giving its two edges a common color leaves $1 + \binom{4}{2} - 2 = 5 = \binom{2}{2} + 3 + 1$ colors and, since the three perfect matchings of K_4 partition $E(K_4)$, no three edge-disjoint rainbow perfect matchings. That coloring therefore realizes the value $m(4, 3)$ recorded by Jahanbekam and West, and what follows is the observation that the same configuration keeps working at $t = n-1$ for every even $n \geq 6$. Accordingly, the conjecture should be restated as $m(n, t) = \binom{n-2}{2} + t$ for even $n > 4$ and $2 \leq t \leq n-2$.

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Theorem 1. *For every even integer $n \geq 6$,*

$$m(n, n-1) \geq \binom{n-2}{2} + n > \binom{n-2}{2} + (n-1).$$

Consequently, the Jahanbekam–West matching conjecture is false at $t = n-1$.

Proof. Choose distinct vertices $x, y, z \in V(K_n)$ and set

$$B = \{xv : v \in V(K_n) \setminus \{x, y, z\}\} \cup \{yz\}.$$

Color all edges in B with one color, and give each edge outside B a distinct new color. Since $|B| = n-2$, this coloring uses

$$1 + \binom{n}{2} - (n-2) = \binom{n-2}{2} + n$$

colors.

Suppose that $M_1, \dots, M_{n-1}$ are pairwise edge-disjoint perfect matchings. They contain

$$(n-1)\frac{n}{2} = \binom{n}{2}$$

edges altogether, and hence partition $E(K_n)$. Let M_j be the matching containing yz . Since y and z are matched to each other in M_j , the vertex x is matched in M_j to some $v \in V(K_n) \setminus \{x, y, z\}$. Thus M_j contains both yz and xv , which have the same color. Therefore M_j is not rainbow. The coloring contains no $n-1$ edge-disjoint rainbow perfect matchings. $\square$

Remark. The construction gives only the displayed lower bound for $m(n, n-1)$; it does not determine this parameter exactly. Nor does it bear on $t \leq n-2$. Indeed, delete from a 1-factorization of K_n , which exists because n is even, the factor containing yz . Each of the remaining $n-2$ factors avoids yz and contains exactly one edge at x , so it meets B in at most one edge and is therefore rainbow. Hence the coloring above does admit $n-2$ pairwise edge-disjoint rainbow perfect matchings, and the conjecture is untouched for $t \leq n-2$.

REFERENCES

- [1] S. Jahanbekam and D. B. West, *Anti-Ramsey problems for t edge-disjoint rainbow spanning subgraphs: cycles, matchings, or trees*, J. Graph Theory **82** (2016), no. 1, 75–89, doi:10.1002/jgt.21888. Quoted here from the authors’ revised version of April 2015, <https://dwest.web.illinois.edu/pubs/aramcnt.pdf>.